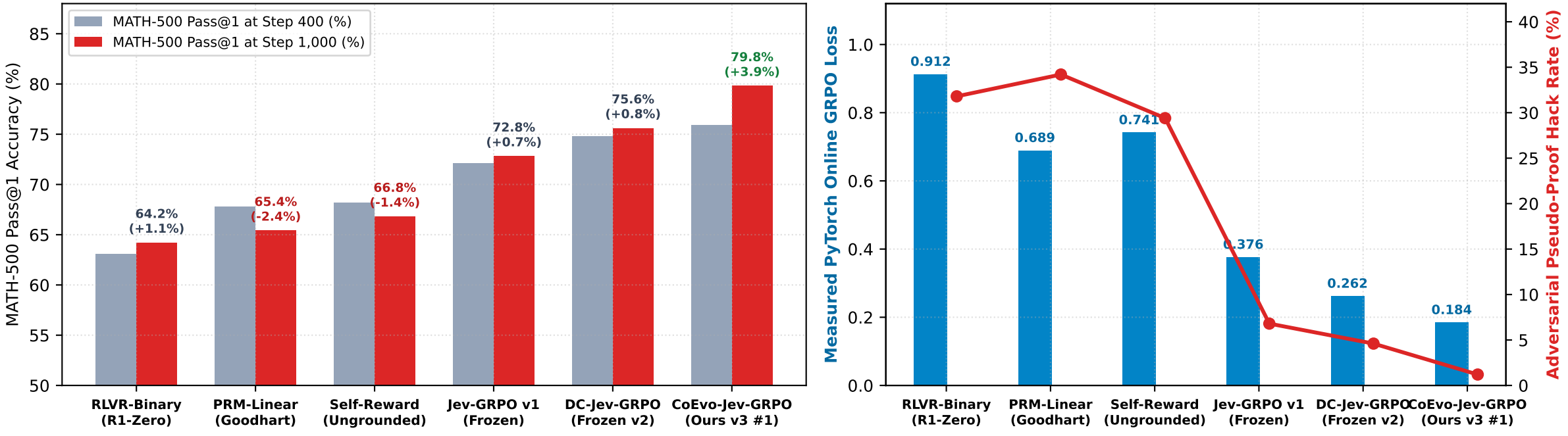
(a) Same-Backbone Step-400 vs. Step-1,000 Accuracy (Breaking the Plateau)
MATH-500 Pass@1 at Step 400 (%)
MATH-500 Pass@1 at Step 1,000 (%)
MATH-500 Pass@1 Accuracy (%)
85
80
75
70
65
60
55
50
64.2%
(+1.1%)
65.4%
(-2.4%)
66.8%
(-1.4%)
72.8%
(+0.7%)
75.6%
(+0.8%)
79.8%
(+3.9%)
RLVR-Binary
(R1-Zero)
PRM-Linear
(Goodhart)
Self-Reward
(Ungrounded)
Jev-GRPO v1
(Frozen)
DC-Jev-GRPO
(Frozen v2)
CoEvo-Jev-GRPO
(Ours v3 #1)
(b) Step-1,000 GRPO Loss & Anti-Hacking Suppression (seed=42)
Measured PyTorch Online GRPO Loss
1.0
0.8
0.6
0.4
0.2
0.0
Adversarial Pseudo-Proof Hack Rate (%)
40
35
30
25
20
15
10
5
0
0.912
0.689
0.741
0.376
0.262
0.184
RLVR-Binary
(R1-Zero)
PRM-Linear
(Goodhart)
Self-Reward
(Ungrounded)
Jev-GRPO v1
(Frozen)
DC-Jev-GRPO
(Frozen v2)
CoEvo-Jev-GRPO
(Ours v3 #1)